

## **Phase 1 – Requirements Analysis**

### **Smart ATM Routing System**

**Prepared for: NBE Internship Project**

Version: 1.0

Status: Ready for Review

#### **1. Project Overview**

Smart ATM Routing System is a web application that helps customers locate the most suitable ATM based on operational status, supported services, cash availability, congestion level, and distance. The system reduces wasted trips by recommending the best ATM rather than simply the nearest one.

#### **2. Problem Statement**

Customers often travel to an ATM only to discover that it is out of service, out of cash, or does not support the required transaction. This wastes time and negatively affects customer satisfaction.

#### **3. Project Objectives**

- Recommend the best ATM for the requested service.
- Reduce unnecessary travel.
- Display real-time ATM information.
- Show ATM details and supported services.
- Allow customers to report ATM issues.
- Integrate navigation using Google Maps.

#### **4. Project Scope**

In Scope

- Search nearby ATMs
- Filter by transaction type
- View ATM details
- View ATM status
- View supported services
- View cash denominations

- View congestion level
- Smart recommendation algorithm
- Navigation to ATM
- Report ATM issues

Out of Scope

- User registration/login
- Admin dashboard
- Employee management
- ATM hardware control
- Online banking transactions

## 5. Stakeholders

- Customer
- National Bank of Egypt
- ATM Monitoring Service
- Google Maps API

## 6. System Actors

Primary Actor

- Customer

Secondary Actors

- ATM Monitoring Service
- Google Maps API

## 7. Functional Requirements

FR-1 Search nearby ATMs.

FR-2 Filter by transaction type.

FR-3 View ATM details.

FR-4 View supported services.

FR-5 View cash denominations.

FR-6 View congestion level.

FR-7 Recommend the best ATM.

FR-8 Open navigation.

FR-9 Report ATM issues.

## **8. Non-Functional Requirements**

- Response time < 2 seconds
- Secure communication (HTTPS)
- Responsive UI
- High availability
- Scalable architecture
- Maintainable layered design

## **9. Business Rules**

BR-1 Only operational ATMs can be recommended.

BR-2 ATM must support the selected transaction.

BR-3 Cash withdrawal requires available cash.

BR-4 Congestion decreases recommendation priority.

BR-5 If no ATM matches, recommend the nearest branch.

BR-6 Customer reports are stored for later verification.

## **10. Assumptions**

- ATMs periodically update their status.
- Cash information is available.
- Customers allow location access.
- Internet connection is available.
- Google Maps API is accessible.

## 11. Constraints

- ASP.NET Core Web API
- SQL Server
- Google Maps API
- Responsive Web Application

## 12. User Stories

US-1 As a customer, I want to find the best nearby ATM.

US-2 As a customer, I want to know which services are available.

US-3 As a customer, I want to view congestion before visiting.

US-4 As a customer, I want navigation to the selected ATM.

US-5 As a customer, I want to report incorrect ATM information.

## 13. Acceptance Criteria

- Search returns recommended ATMs.
- Recommendation algorithm works.
- ATM details are displayed.
- Navigation opens correctly.
- Reports are stored successfully.
- APIs return valid data.

## 14. Phase 1 Deliverables

✓ Approved Requirements

✓ Final Project Scope

✓ Functional Requirements

✓ Non-Functional Requirements

✓ Business Rules

✓ Actors

✓ User Stories

✓ Acceptance Criteria

Next Phase: Database Design (ERD & SQL Server).